SCTR's
Pune Institute of Computer Technology, Pune - 411043.
Department of Artificial Intelligence (AI) & Data Science (DS)

In-House Internship on AyurSutra

Deep Learning and Computer Vision for **Pharmaceutical Authentication**

Under the Guidance of

Dr. Shweta Dharmadhikari

Presented by:

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Domain: AI / ML · Healthcare · Ayurveda

Project Overview & Motivation

Topic Details

Topic:
AyurSutra — Deep Learning & CV for Pharmaceutical Authentication

Mentor:
Dr. Shweta Dharmadhikari

Domain:
AI / ML in Healthcare (Ayurveda)

Motivation

Digital Transformation
Need to digitize Ayurvedic healthcare while preserving its holistic nature.

Error Reduction
Manual therapy scheduling and prescription validation are prone to human error.

Unified Platform
Lack of existing AI platforms for automating Ayurvedic therapy planning and formulation authentication.

Aim & Objectives

AIM
To design an intelligent framework automating Ayurvedic therapy scheduling and authenticating herbal formulations using Deep Learning and Computer Vision

Intelligent Scheduling

Automate therapy appointments based on patient Dosha and diagnosis.
Priority-based scheduling with starvation prevention — higher-priority appointments are allocated first, while remaining sessions follow system allocation.

Authentication

Employ OCR and CNN to validate herbal medicine labels and formulations.
Identify Panchakarma type and generate complete treatment plans, reducing doctor workload.

Interpretation

Integrate NLP to interpret multilingual prescriptions and classical Ayurvedic terminology, enabling cross-lingual prescription understanding.

Literature Review Summary

Author / Year	Paper Title	Method	Key Findings
Shaumya & Kumar (2025)	AI in Ayurveda: Systematic Review	ML & DL Survey	NLP + CNN key drivers; 60+ papers reviewed
IRJET (2025)	AI-Driven Prescription & Health Insight	CNN + OCR + NLP	90% accuracy in text-image prescription validation
Manjula et al. (2025)	Ayurvedic Diagnosis using ML	Random Forest & Decision Tree	87% classification accuracy on herbal datasets
STM Journals (2025)	AI in Ayurvedic Healthcare: Survey	Comparative AI study	Highlights integration challenges & lack of multilingual data
Bibliomed (2025)	Transforming Ayurveda through AI	Knowledge Graph + Text Mining	2000+ herb-dosha relations extracted from ancient texts

System Architecture

Tech Stack: Firebase | Node.js | Flask | TensorFlow | React Native | Python

Patient Portal

- Onboarding (eKYC)
- Dosha Quiz & Analysis
- Appointment Booking
- Priority Scheduling
- Herbal Remedy Advisor
- Diet Planner
- Report Analyser
- BPM Monitor (Webcam)

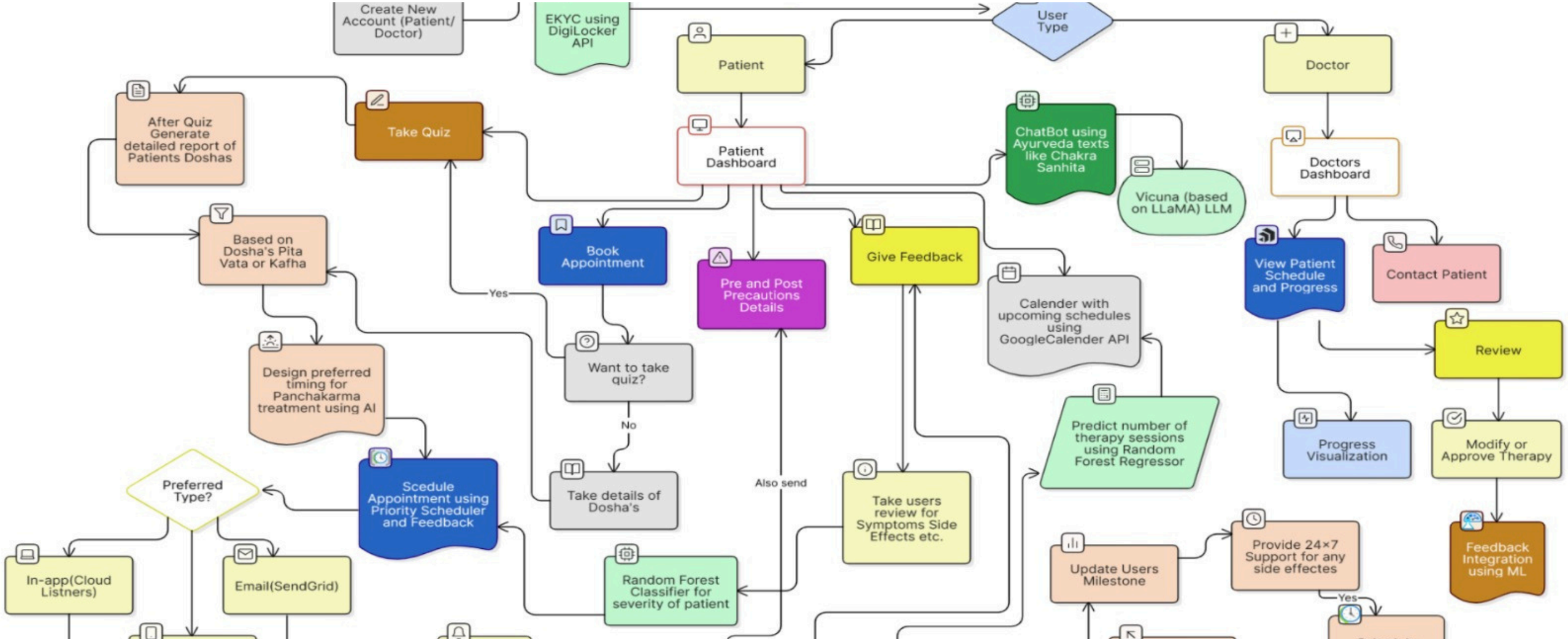
Core AI Modules

- Smart Scheduler — Random Forest Regressor
- Digital Vaidya — LLaMA/Vicuna LLM Chatbot
- Dosha Classifier — Random Forest Classifier
- OCR + CNN — Medicine Label Authentication
- NLP Engine — Multilingual Prescription Parsing
- Panchakarma Planner — Auto Treatment Plan

Doctor Dashboard

- Priority Queue Management
- Therapy Approval System
- Patient Progress Tracking
- Separate Secure Login (Role-based Auth)
- Treatment Plan Review

System Architecture Diagram



Flow Diagram

[Insert Flow Diagram Here]

Model / Feature Diagram

[Insert Model / Feature Diagram Here]

Challenges

Data Scarcity

Major lack of digitized, annotated datasets for Ayurvedic prescriptions. Classical texts are predominantly in Sanskrit and regional scripts, making automated data collection significantly difficult.

Lack of Integration

Existing digital health apps monitor therapy but lack deep Ayurvedic integration (Dosha / Prakriti assessment). No platform currently bridges modern scheduling with authentic Ayurvedic logic.

Linguistic Barrier

No specific NLP models are tailored for interpreting multilingual Ayurvedic terminologies. Prescriptions span Sanskrit, Hindi, Marathi, and English — requiring custom multilingual models.

Current Status of the Project

Phase 1
Research & Design

● COMPLETED

Literature Survey and Gap Analysis — Completed

System Architecture and Data Flow Diagrams — Finalized

Phase 2
Model Development

● COMPLETED

Dosha Classifier: Random Forest Classifier — Implemented for patient severity analysis

Session Predictor: Random Forest Regressor — Trained for therapy session estimation

Priority Scheduling Algorithm — Implemented with starvation prevention

Panchakarma Type Identifier — Algorithm integrated & tested

Phase 3
Prototype

● COMPLETED

Patient and Doctor Dashboards (UI/UX) — Developed

Separate role-based secure logins for Doctors and Patients — Implemented

Features Integrated: Herbal Remedy, Diet Planner, Report Analyser, BPM Monitor via Webcam

Conclusion & Future Scope

Conclusion

AyurSutra bridges ancient Ayurvedic wisdom with modern Artificial Intelligence to enhance therapy accuracy and scheduling efficiency.

Impact: Reduces manual errors, optimises resource allocation, and ensures genuine medicine validation through OCR and CNN-based authentication.

Thank You

AyurSutra — Bridging Ancient Wisdom with Modern Intelligence

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